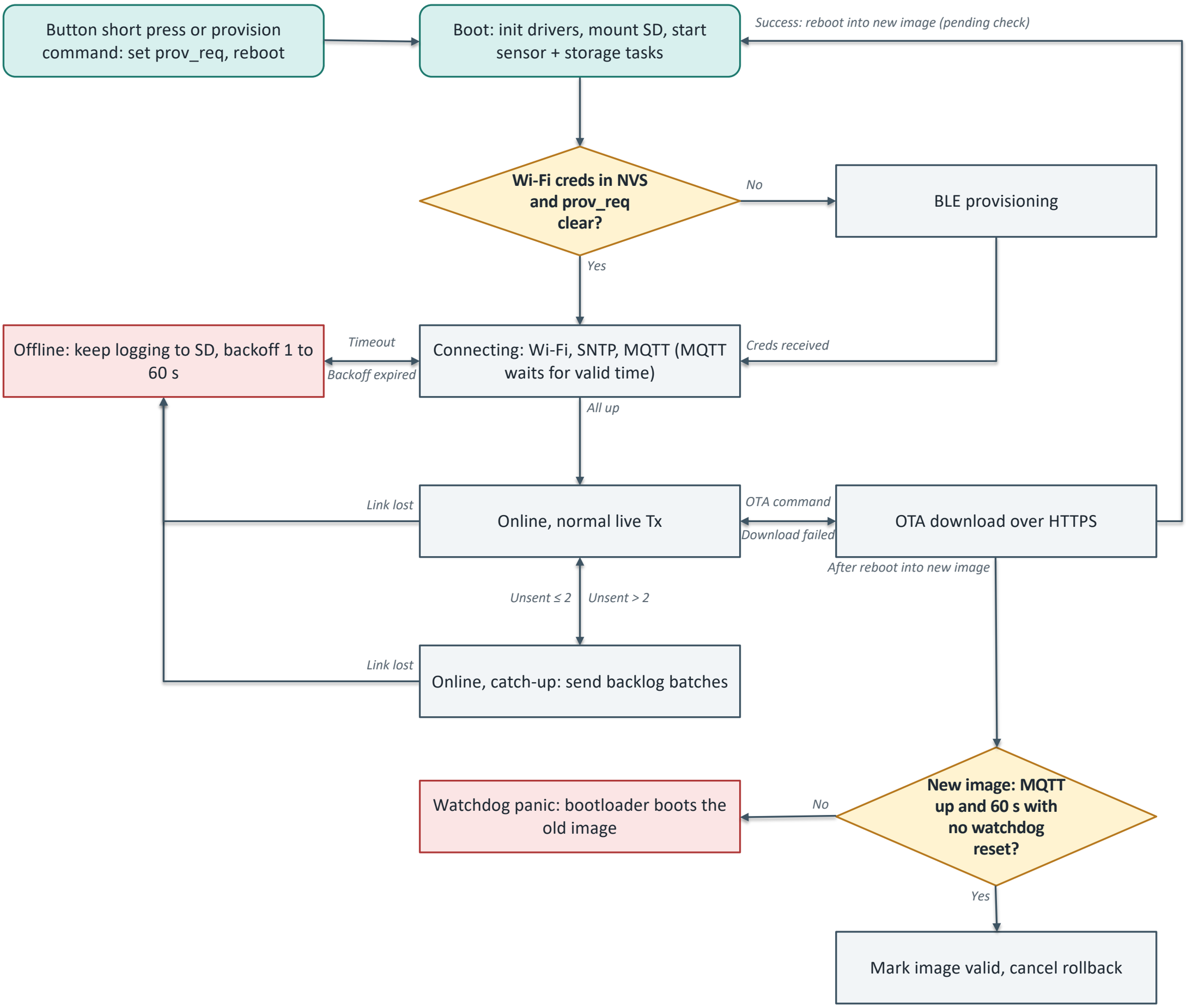
Top-level state machine (connectivity)

Sampling and storage run independently in every state

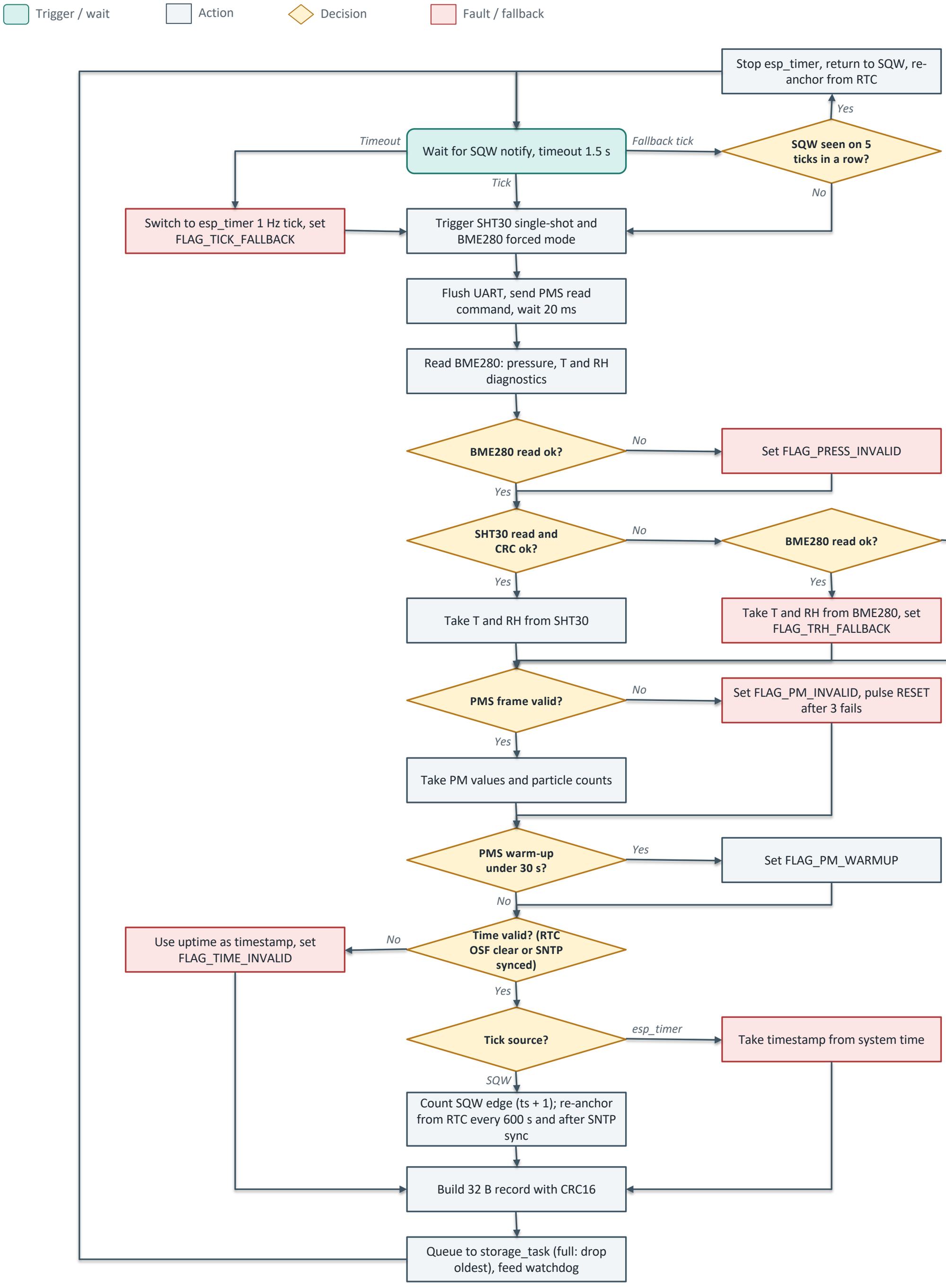
Trigger / wait | Action | Decision | Fault / fallback



Note: sensor_task and storage_task run in ALL states at 1 Hz, SD first. Arrows with two heads carry both transitions (labels above/right and below/left)

sensor_task

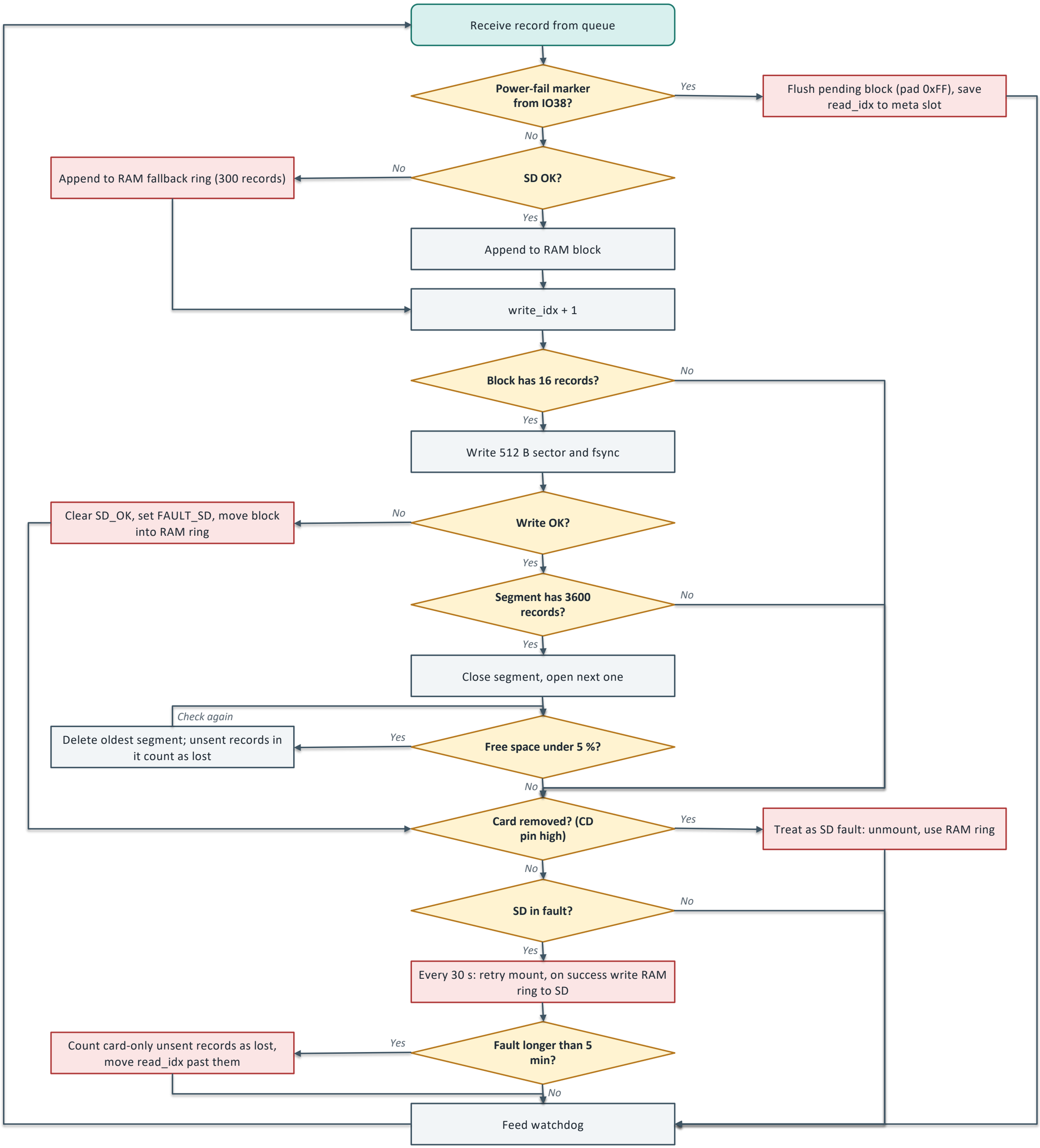
Core 1, triggered by DS3231 SQW (1 Hz)



Note: PMS frame check: header 0x42 0x4D, length 28, checksum, PM1 ≤ PM2.5 ≤ PM10. Timestamps count SQW edges and are re-anchored from the RTC.

storage_task

Core 1, consumes the record queue, writes 512 B sectors

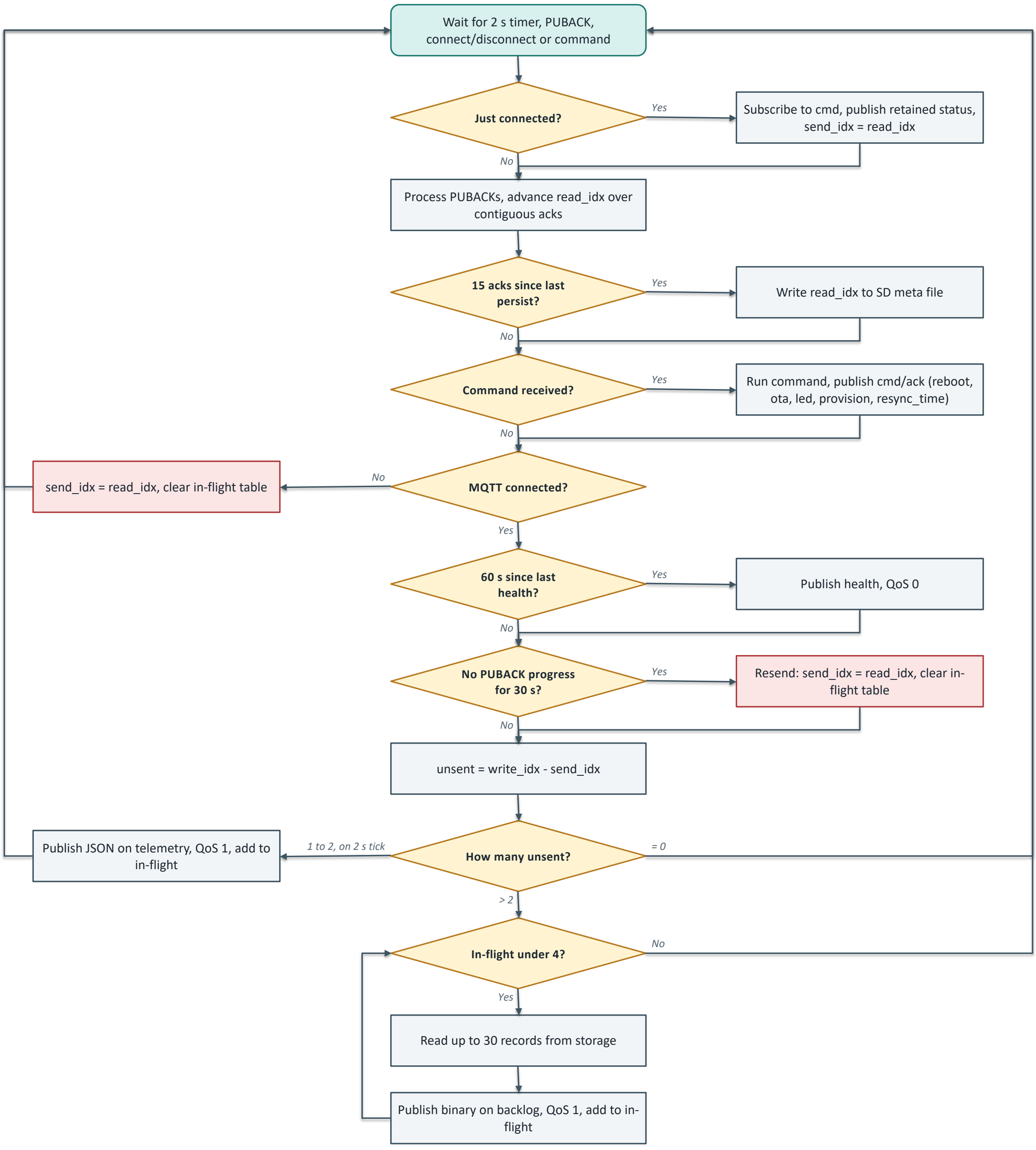


Note: 1 segment = 3600 records x 32 B = 115,200 B = 225 sectors. Power-loss window: max 16 s, closed by the IO38 power-fail flush.

publish_task

Core 0, single FIFO with read_idx (acked) and send_idx (next to send)

Trigger / wait | Action | Decision | Fault / fallback



Note: On disconnect, unacked records are resent. Backend dedups on (device_id, seq). Only this task publishes.